

Global Industrials

ABB/Legrand CMDs and the 800V DC debate: Asking the right questions

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Summary

ABB (Sep 24) and **Legrand** (Sep 29) are hosting CMD focused on datacenters and the transition to 800V DC. In this report we review the expectations and provide an updated primer on 800V DC, including SSTs and MV UPS.

It's time to move on. The initial phase of the architecture transition is now understood and extensively debated, including its negative relative impact on Legrand.

Could industry profitability be lower in 2030 than today? The technology transition in the longer term could negatively impact industry profitability. While current leading vendors should master the transition successfully, the market structure may be less favorable.

We continue to see near term upside to growth: Order trends for power and the recent NVIDIA 2027 guidance imply that datacenter estimates across the electrical equipment suppliers have upside. Consensus assumes that the non-datacenter growth differential between ABB vs Legrand widens further.

ABB playing offence: The company is well positioned for 800V DC, even if it also faces transition challenges. We expect a confident presentation. **Legrand fighting back:** The company could lose 2-3% of DC revenue growth p.a. over a number of years. This is digestible in a strong double digit growing market, and we expect the company to highlight product innovations to benefit from 800V DC. Contrary to others, current profitability does not imply its overearning, a differentiator.

Portfolios are dynamic, the 800V DC transition is fluid and not set in stone. We caution to credit companies with % TAM shares, based on a prototype or start-up acquisitions vs other that have yet to make a move.

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Upcoming events to refocus investors on 800V DC transition

Scheduled events by ABB and Legrand will likely refocus investor debate on the technology transition triggered by the move to higher-voltage direct current technologies needed to power the next generations of GPUs. We believe the current debate is overly focused on the near term transition to the 800V sidecar setup, negatively impacting Legrand's rack PDU business and often based on matching current offerings to an expected future addressable market, forgetting that portfolios are dynamic. The real question, beyond the overall datacenter capex outlook, is how the likely multi-stage transition over the next 5 years is shifting profit pools, competitive dynamics and industry structures.

ABB (Sep 24, [Webcast on DC datacenters](#)) – playing offence

- We expect the Electrification Management team to focus on ABB's strength in technology and solution offering in direct current technologies. The company's industry-first medium voltage UPS offering, low exposure to traditional UPS and direct current technology expertise stemming from its Marine Electrification and EV-charger business position it well. It also owns a stake in the solid state transformer company DG Matrix.
- ABB is well positioned in Medium Voltage but also faces a transition at the low voltage electrical level, a very substantial part of its offering, and is less well positioned closer to the server and in cooling, areas that benefit overtime from fast replacement cycles, when compared to the power infrastructure that can be reused for future chip generations. We don't expect the event to include information on current trading or discussion of financial targets.

Legrand ([CMD in Singapore, Sep 29, in conjunction with Datacenter World](#)) – addressing investor concerns

- This is a highly anticipated event where we expect Legrand to "fight back" on the narrative that it is a relative loser from the technology transition, as reflected in the derating seen this year, and several "feisty" debates on quarterly conference calls with analysts. We expect to see a lot of detail around new product introductions for 800V DC.
- Investor feedback point to on balance long position by hedge funds into the event, supported by consensus sales growth expectations and company targets that appear too low if the pace in datacenter growth continues.
- At its last CMD in September 2024, Legrand set a €12-15bn sales target for 2030, including 3-5% organic sales CAGR and 3-5% annual contribution from M&A while maintaining a 20% adj. EBIT margin. The company stated at the H1 2025 results that it has increased confidence that it could reach the upper end of the target. With the FY 2025 results it formally moved to the €15bn sales target by 2030 and aimed for margins to stay >20%. Bloomberg consensus looks for 2026 revenues of €11bn. Consensus does not include acquisitions yet to be announced and hence the €15bn 2030 consensus number substantially understates investor expectations for total growth. Note that the comparative data we have shown in Figure 1 does not include any M&A contribution either. Therefore as a point of principle, we would refer to the organic growth targets as opposed to interpretations of 'headline reported consensus'.

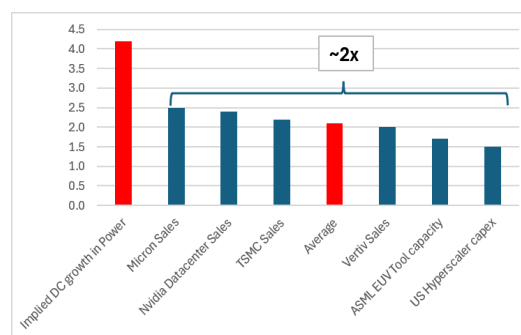
- Consensus looks for organic growth of 10% in 2026, stepping down 8% in 2027 and then in a linear trajectory to 5% by 2030 with an adj. EBIT margin of 22%. We believe that the growth trajectory may be too conservative, based on the assumptions shown in Figures 1 and 2. We have previously shown that expectations for a number of different parts of the Electrification ecosystem (NVIDIA, Micron, TSMC, Vertiv) imply revenue growth multiplier 2.1x between 2026 and 2030 and this implies a growth CAGR 20%. In a Base Case applying this to Legrand, and assuming 4% growth in the non-DC portion, this implies a 9.8% organic growth CAGR versus ~3-5%. It would also imply a LSD upgrade to reported estimates alone; even prior to accounting for M&A. The ~10% organic is a potential operational Base Case, though note we would be surprised if Legrand formally underwrites that.
- A more Conservative scenario would be a ~15% Datacenter CAGR and ~2% implied organic on the non-DC businesses, getting the group organic to 6.6%. This is still above the ~3-5% range. On this basis, we would not be surprised to see a >2ppt upgrade to organic growth targets, eg. into a ~5-7% band. However, whether this would be enough to "excite the market" is a separate question. As we review some of the recent market commentary, we are a little concerned that expectations (and positioning) becomes exaggerated. Note that Legrand is an understated company with a consistent history of under-promising and over-delivering. Thankfully, this company does not get sucked into the "showbiz element" that some larger peers appear to subscribe to.
- We have not really touched on operating margin. We see market expectations in 2030 approaching the ~22% level. This is historically out of character for Legrand; but understandable given operating leverage in the Datacenter portion. Against this, Legrand will be re-investing in its business at a time of technology transition (800v DC). In our view, any margin guide >21% would be a positive surprise. Legrand's target also includes the likely dilutive effect of M&A, which is not reflected in consensus as analysts don't model future potential acquisitions.
- We also look for incremental details on a number of acquisitions completed over the past year, including some that while increasing the datacenter exposure, may not get recognized by investors as typical Legrand businesses. Legrand made a number of meaningful acquisitions that further shifts its exposure to Datacenters, particularly in the US, reducing the contribution of its Rack PDU business. (Gritz Industries, SRS Power, Keydak, TES, Avtron Power). The company has very actively shaped its datacenter portfolio over the past 10 years from a minor business to a leading position with revenues now accounting for one third of group sales. These acquisitions, often at attractive valuations, demonstrate management's ability to build and shape a business over time. Some of the acquisitions, like the load bank business Avtron, have different characteristics and it will be interesting to hear how they fit into the Legrand business model.
- The basis for our Datacenter growth CAGRs are shown in Figure 2. This shows consensus expectation for a number of Electrification datapoints between 2026 and 2030. We have updated for NVIDIA's latest 2027 ~70% sales forecast, resulting in a 2.4x multiplier over that timeframe. Micron is at 2.4x sales increase between 2026-30, Vertiv 2.0x and ASML's EUV tool capacity is 1.7x. The average multiplier for the Electrification ecosystem related to datacenters is 2.1x over four years - ie. a doubling.

Figure 1: Base and Conservative Scenarios – Oxcap believes that Legrand can grow at 9.8% through 2030. However, we expect growth targets to be upgraded more conservatively, from ~3-5% to ~5%-7% band pre-M&A

Base	2026	New 2030	2030 cons	Diff	CAGR	Multiplier
Datacenter	3,687	7,645	-	-	20.0%	2.1
Non-Datacenter	7,157	8,372	-	-	4.0%	1.2
Total	11,012	16,017	15,493	3.4%	9.8%	
EBIT	2,346	3,444	3,362	2.4%	-	
Margin	21.3%	21.5%	21.7%	-	-	
Conservative	2026	New 2030	2030 cons	Diff	CAGR	Multiplier
Datacenter	3,687	6,448	-	-	15.0%	1.7
Non-Datacenter	7,157	7,747	-	-	2.0%	1.1
Total	11,012	14,195	15,493	-8.4%	6.6%	-
EBIT	2,346	2,981	3,362	-11.3%	-	-
Margin	21.3%	21.0%	21.7%	-	-	-

Source: Oxcap Analytics. Note that reported consensus is pre-M&A assumptions

Figure 2: Electrification Ecosystem – the average 2026-30 sales increase is ~2.1x which implies a ~20% CAGR. Applying the same math to Legrand and a 4% non-DC growth generates a 9.8% sales CAGR



Source: Oxcap Analytics, based on company data

Figure 3: Legrand – Consensus Estimates vs Company Targets

Metric	2025A	2026E	2026 Target	2027E	2028E	2029E	2030E	25-30 CAGR	Mid-Term Target (2030)
Sales (EUR mn)	9,481	11,012		12,305	13,334	14,495	15,316	10.1%	c.€15bn sales
Total Growth %	9.6%	16.1%	+16% to +19% (ex-FX)	11.9%	8.6%	8.0%	6.8%	10.3% (avg)	close to +10% avg (ex-FX)
Organic Growth %	7.7%	9.6%	+8% to +10%	9.0%	6.6%	5.7%	4.7%	7.1%	+3% to +5% CAGR (old, 2024 CMD)
Adj. Operating Margin %	21.4%	21.3%	20.5% to 21.0%	21.5%	21.7%	21.6%	21.6%		Average above 20%
EPS Growth %	8.6%	10.0%		15.7%	10.9%	8.6%	6.0%		
FCF (EUR mn)	1,331	1,482		1,782	1,995	2,142	2,304	11.6%	Nearly €10bn cumulative over 25-30; avg 13% to 15% of sales
FCF (% of sales)	14.0%	13.5%	13% to 15% of sales	14.5%	15.0%	14.8%	15.0%		
Capex (EUR mn)	221	252		275	293	318	332		
Capex (% of sales)	2.3%	2.3%		2.2%	2.2%	2.2%	2.2%		Avg 3% to 3.5% of sales

Source: Visible Alpha

Market expectations

Neither Legrand nor ABB provide detailed revenue disclosure about their datacenter business. Legrand commented that datacenter revenues grew at more than 30% in H1 2026 (like for like) while also announcing a number of acquisitions in the datacenter electrification space. Hence the table below represent rough estimates. We have assumed uniform growth of the underlying datacenter business for arguments sake to calculate the growth of the non-datacenter business implied by current group revenue consensus (for Legrand) and segment consensus (for ABB Electrification). For Legrand, the non-datacenter business is expected to decline by 3-4% per year to 2028, which compares to ABB growing at 9%. ABB Electrification has a much more attractive end-market mix for the current business environment, but Legrand's consensus estimates appear to have more relative upside in the medium term should construction activity normalize. Legrand's ~41% exposure to non-residential construction (2025 revenues) is skewed towards weaker verticals like commercial and office when compared to ABB, and the industry overall. ABB Electrification derives ~20% of revenues related to Power and Distribution and 25% from Industry, in addition to its 17% exposure to Datacenters.

Figure 4: Legrand in Euro million

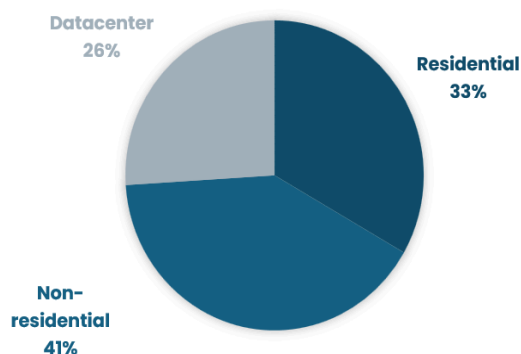
	2024	2025	2026E	2027E	2028E
Revenues	8,649	9,481	11,017	12,191	13,145
Organic growth		9.8%	7.8%	6.7%	5.7%
Datacenter revenue	1,750	2,465	3,856	5,356	6,427
Acquisitions		NA	500	150	0
Growth		NA	56%	39%	20%
Organic growth		40%	40%	35%	20%
Share		26%	35%	44%	49%
Implied organic growth ex DC		1.5%	-1.6%	-5.6%	-3.1%

Source: Company data, Bloomberg as of Sept 21, 2026 for revenues. Oxcap Analytics.

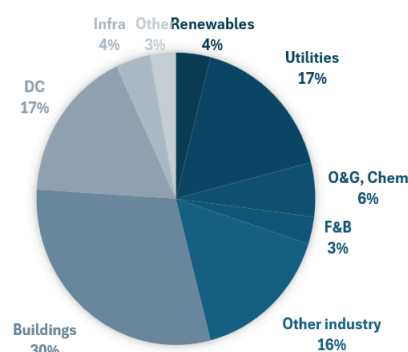
Figure 5: ABB Electrification in \$ million

	2024	2025	2026E	2027E	2028E
Revenues	15,448	17,357	20,868	23,834	26,688
Organic growth		9.0%	11.0%	14.9%	11.4%
Datacenter revenue	1,854	2,951	4,131	5,577	6,692
Acquisitions		NA	0	0	0
Growth		59%	40%	35%	20%
Organic growth		59%	40%	35%	20%
Share		12%	20%	23%	25%
Implied organic growth ex DC		3.9%	12.0%	8.0%	6.7%

Source: Company data, Bloomberg as of Sept 21, 2026 for revenues, Oxcap Analytics.

Figure 6: Legrand end-market breakdown 2025 revenues


Source: Company data.

Figure 7: ABB Electrification end-market breakdown 2025 revenues


Source: Company data.

The key debates on 800V DC

- The initial transition, starting in 2027, to a 800V DC side-car design is well understood and visible. What comes after, and at what pace, is still somewhat unclear. The big question is whether the architecture will use a Solid State Transformer or use traditional transformers with rectifiers is TBC. While the traditional low voltage UPS likely disappears in AI datacenters, it is unclear what share of the value migrates upwards to a MV UPS.
- The overall \$/MW opportunity for electrical vendors does not change meaningfully, but competition and market fragmentation likely does. In Phase 1, the revenue opportunity increases for most, which together with likely stronger DC capex could result in further upside to consensus estimates in the short term.
- What share of the datacenter market adopt a 800V DC architecture, and by when? Vendor expectations vary widely (often linked to whether they are a beneficiary or not). Is 800V DC necessary for interference? What is the non-NVIDIA chip ecosystem doing?

- Legrand is the only vendor with a negative net value \$/MW impact in Phase 1 of the transition, with other vendors generally benefiting. However, portfolios are not static and vendors adapt.
- To what degree will the technology change allow new entrants to the market? Will current market leaders continue to dominate, because customers want to deal with broad-based, established vendors with solution capabilities?
- How will vendors deal with introducing new technologies, higher R&D and ramp down/ramp up of new products, with often short lifespans (for example sidecar), in an environment where execution is already a key topic (as with Vertiv in Q2), given the fast growth of demand.
- What happens to profit pools? The AI datacenter opportunity was incredibly favorable to existing market leaders in electrical products. After long periods of low growth, the supplier universe consolidated with strong market shares in key categories for the leaders, like Eaton and Schneider in UPS, Siemens, ABB and Schneider in Medium Voltage or Legrand in rack PDUs and some other installation products. Initially, we will likely see a more fragmented vendor set-up for new products such as the sidecar or later SSTs. Over the past five years, vendor profitability had improved strongly due to the high demand and less price focused datacenter customers, with medium voltage margins improving from low teens to the 20s and low voltage business from the high teens to the mid or high 20s.
- Will the technology transition open the door to Asian electronics suppliers and contract manufacturers more broadly? Companies like Delta Electronics, LiteOn or Flex are well positioned given their electronics and direct current capabilities. Will hyperscalers develop more of their own power technology, outsourcing production to contract manufactures, rather than buying standard products from the current market leaders? China is implementing DC technologies quicker than the West, but its vendors may miss out on the US opportunity given tariffs and Presidential executive orders.

Overview of the 800V DC technology transition

Why is the datacenter architecture changing and what it means

- **Higher-voltage DC is becoming a physics-led requirement, not a design choice:** The move towards a higher-voltage direct current architecture for AI datacenter power supply over the next five years is increasingly being driven by physics, given the power densities expected in next-generation compute. 800V DC — a higher-voltage “low-voltage” architecture rather than a high-voltage system — appears set to become the preferred architecture, although even higher voltages may be required over the longer term.
- **The objective is to power more compute within a denser layout with lower infrastructure cost:** The ultimate goal of these new architectures is to enable more powerful compute capacity in a dense setting to be powered with lower capex and opex in the supporting electrical infrastructure. This is generally negative for electrical supplier growth relative to the underlying growth in gigawatts of compute, as each MW of compute may require less traditional electrical equipment than would have been the case with today's architecture.
- **The market shifts from grey-space AC equipment towards white-space power electronics:** Under an 800V DC architecture, the electrical vendor opportunity at the medium-voltage level remains largely intact. However, the grey-space low-voltage market transitions from AC to DC, reducing the role of traditional LV UPS, switchgear, transformers and PDUs to auxiliary facility-level power loads and cooling. At the same time, the share of spending captured by white-space power electronics increases.
- **The total opportunity per MW changes only modestly, but the mix changes significantly:** According to the SemiAnalysis framework, the addressable market for electrical suppliers falls from c.\$3.35m per MW of compute in the baseline scenario, excluding generators, to c.\$3.05m per MW in Phase 4. This reduction is modest relative to the expected underlying growth in compute demand. However, the composition of spend changes dramatically: c.\$1.25m of the Phase 4 opportunity is represented by the emerging solid-state transformer solution, a new technology category without established market leaders, where start-ups and transformer manufacturers such as GE Vernova may compete with traditional DC electrical equipment vendors and the low voltage AC incumbents. The addressable market in Phase 1 expands, with only rack PDUs impacted.

Technology transitions matter

To put it in a simplistic way, traditionally, datacenters use standard AC low voltage electrical products that are widely available due to their use in industrial and commercial applications. Electricians and installers are familiar with the products, and standards clearly defined. Over the last few years, the datacenter market has grown to proportions where it can have its own designs and standards, with NVIDIA and the Open Compute Project (OCP) leading the standardisation and design of reference architectures. Over the past 5 years, LV electrical suppliers increased capacities in their manufacturing to supply the new market. If these products transition to new, dedicated DC solutions, competition may increase, and production capacities need to be adapted with R&D increasing. The transition may increase opportunities for companies like Delta and Flex with their strong contract manufacturing capabilities.

- **Pricing conditions may change:** To date, the AI datacenter power opportunity has largely accrued to established electrical suppliers, often through their existing product portfolios in relatively consolidated markets. Strong datacenter demand has tightened capacity and supported pricing power, benefiting their broader medium- and low-voltage power businesses and contributing to material margin expansion. As datacenters transition towards DC architectures for the lower-voltage power system, pricing conditions in traditional AC markets could begin to change.
- **Shift from electromechanical to power electronics - margin questions:** The shift towards higher power-electronics content, and away from more traditional electro-mechanical equipment, may also be negative for margin mix. It could increase the share of outsourced supply and expose incumbents to greater competition, particularly from Asian vendors with strong positions in power electronics.
- **Growth effects not visible until 2028-29:** In the near term, Phase 1 of the transition appears broadly additive to the vendor opportunity, with the exception of the rack PDU market, only relevant to Legrand in terms of size of exposure, while underlying growth in compute demand is expected to remain strong. As a result, we would expect the technology-driven impact on demand and vendor growth to become more visible only from around 2028 or 2029 onward. We continue to see upside to near term growth expectations for the vendors.
- **Hyperscale capex - important variable:** However, should this technology transition coincides with a moderation in hyperscalers capex growth towards the end of the decade, the impact of the technology transition could become more material for individual vendor growth rates.

Oxcap's points of differentiation on the 800V DC transition

- We expect the transition to be more fluid than the clearly defined four-phase framework outlined by SemiAnalysis, which is heavily anchored around the NVIDIA roadmap. ASIC-based chip designs, as well as architectures from AMD and Chinese vendors, may follow different paths. Datacenter power architectures are already heterogeneous across hyperscalers and other operators, and we expect this to remain the case.
- While it is clear that the traditional AC low-voltage opportunity is likely to shrink materially, DC architectures will also require control, protection and monitoring. This could create additional opportunities that may not be fully captured in the SemiAnalysis framework. Again, we believe that AC and DC outcomes are not binary.
- Technology position is only one factor in vendor selection. The ability to manufacture at scale, maintain consistently high product quality and meet tight delivery schedules is critical. These capabilities, combined with close customer relationships, represent an important advantage for incumbent electrical suppliers, particularly relative to new entrants. We expect the current marked leaders to transition successfully, including with the help of M&A.
- Hyperscalers will also want to maintain a diversified supplier base and are unlikely to push a major technology transition before a sufficiently broad set of qualified vendors are ready. The portfolios of electrical suppliers are not static, and we expect further R&D investments, M&A, partnerships and technology licensing as vendors position themselves for the evolving roadmap. We therefore disagree with the idea that there are already clearly defined winners and losers, particularly given that the technology roadmap beyond Phase 1 is not yet set in stone.

When analysing the datacenter technology outlook, Oxcap applies a multi-layered framework:

1. **Power Consumption vs. Unit Volume:** A key question is whether equipment demand is driven primarily by power consumed or by the number of physical units deployed. As rack power density increases, the number of racks – and therefore the number of rack-level components – grows more slowly than the power required to operate the system. This favours categories that scale with megawatts: power generation, medium-voltage equipment, secure power and cooling. This contrasts with categories that scale more closely with rack count: cabinets, cable management and certain types of PDUs.
2. **Equipment Lifespan:** An important differentiator, as it determines the balance between replacement capex and growth capex. Transmission equipment and large transformers can remain in service for 40 years or more; power generation equipment typically has a 20–30-year useful life; broader electrical equipment may last 10–15 years; while equipment closest to the server – including parts of the cooling and rack-level power chain – may be replaced or redesigned with each compute upgrade cycle, potentially every 4–6 years. As the current breakneck pace of AI capex eventually normalises, replacement-cycle exposure should become a more important driver of relative growth.

3. **Technology Transition:** Investors need to assess how architectural shifts, such as 800V DC, liquid cooling and higher rack densities, affect demand by product category and alter the competitive landscape within each category. Mature, oligopolistic markets often deliver the highest margins and most stable returns. By contrast, technology transitions can open the door to new entrants, shift value pools, and require higher upfront R&D, qualification and product ramp costs before margins normalise. The change from electromechanical content toward electronics and semiconductor-based technologies can have negative implications for existing vendors, also in terms of the profitability profile. Electronics technologies often experience faster price deflation, a higher share of outsourced components and a more fragmented competitive landscape, with Asian suppliers potentially playing a much larger role.

Vendor impact – our assessment

SemiAnalysis concludes that **Delta Electronics** is best positioned for the coming design changes. We agree with this assessment. Delta is a leader in power electronics, has strong DC capabilities and has less to lose from disruption to the traditional AC architecture.

We also view **Vertiv** as well positioned, given its lower relative exposure to categories most negatively affected by the transition. This is supported by the large share of cooling in its business mix and the group's position in electronics-rich infrastructure.

ABB and **Siemens** have a relatively high share of datacenter exposure in medium-voltage products, which should not be materially affected by the transition, and a lower share of exposure to low voltage UPS, a category that could be phased out over time. In Siemens' case there is no UPS exposure. ABB also has a strong position in DC technologies, in our view, supported by its heritage in DC applications such as Marine and EV charging. However, we would also highlight the potentially underestimated exposure of both ABB and Siemens to general low-voltage electrical power solutions in datacenters, parts of which may either disappear or need to transition from AC to DC.

Schneider Electric and **Eaton** are less favorably positioned than Vertiv and Delta for the 800V DC transition. UPS / Secure Power Solutions are central to the datacenter offering of both Schneider and Eaton, accounting for around one-third of their datacenter sales, and could be at risk as architectures evolve. That said, both companies are likely to become key suppliers of the incremental DC architecture, including HVDC power racks, battery racks and related infrastructure. Eaton's recent acquisitions of Boyd, in cooling, and Resilient Power Systems, in solid-state transformers, also changed its exposure map and should be considered in any assessment of its future positioning.

NVIDIA's own 800V DC materials ([Link](#)) list ABB, Eaton, Schneider Electric, Siemens and Vertiv as electrical ecosystem partners, but do not appear to include Legrand in the same 800V DC partner roster. This likely reflects a difference in offering: Legrand provides a set of important components, including rack PDUs, busway, cabinets and connectivity, but is less clearly positioned as an end-to-end electrical systems supplier in the same way as some of its peers.

When assessing the risks to **Legrand**, it is also important to recognise that the rack PDU concern is already discussed among investors. Legrand has built a highly successful datacenter business over the past 8–10 years, supported by M&A, and we would expect the company to adapt its portfolio as the market evolves, including through further acquisitions. In addition, Legrand's product categories have not benefited from the same level of pricing and margin expansion seen in areas such as medium voltage, and likely UPS, where tight supply-demand conditions have lifted profitability for some peers well above historical levels. Those profit pools may be more exposed if market conditions normalise.

While the share of revenues affected by the design transition is highest at Legrand, the exposure is also material for ABB, Siemens, Schneider Electric and Eaton, and to a lesser degree Vertiv. Although cooling is not the focus of this report, that market is also undergoing significant change, particularly with the move towards direct-to-chip liquid cooling. This creates risks for Vertiv's traditional cooling offering, but also incremental opportunities in emerging cooling technologies for Vertiv, Legrand, Eaton and Schneider Electric. ABB and Siemens have limited, if any, exposure to cooling. The currently envisaged two-phase liquid cooling architecture and the use of CDUs could themselves be challenged over time by single-phase liquid cooling solutions that do not require a CDU or new cooling technologies (immersion cooling for example) while traditional AC-powered CDUs may be replaced by DC-powered CDUs.

Vendor exposure map

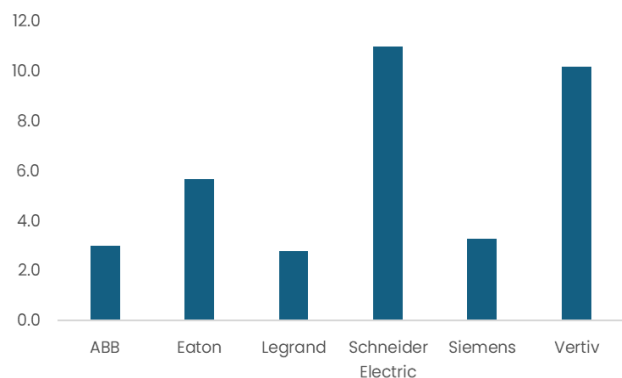
Company disclosure on their datacenter revenues and offering by type is limited. We encourage companies to step up disclosure, now that for many, datacenters are not just the fastest growing market, but also represent a sizeable share of revenues. As the technology transitions, and the break-neck speed of growth eventually starts to normalize, investors will look for more information. The combination of the usual quarterly volatility with investor uncertainty around the technology roadmap may well result in more volatile share prices. We have already seen this in the last two years, where companies are disproportionately punished, or rewarded, on data where there is too much guesswork and backing out. The situation could become far worse, if growth normalizes.

The chart below represents our best effort to split 2025 revenues of the vendors under coverage into standardised categories. Schneider, Legrand, Siemens and Vertiv provide some disclosure by offering but each use their own categories, often bundling groups together. Hence Figure 10 below likely has inconsistencies and mistakes and we are keen to get feedback.

We apply the following breakdown:

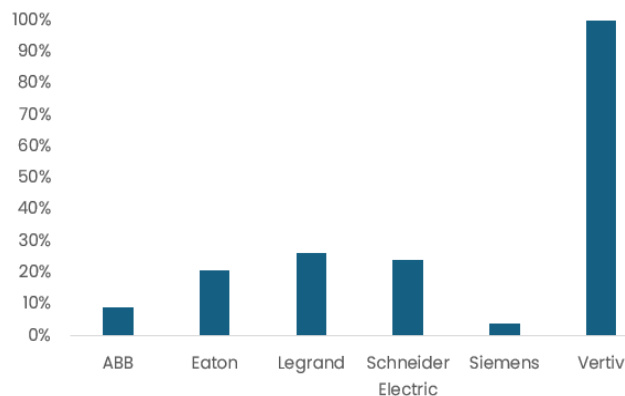
- **Medium Voltage:** Medium voltage switchgear and transformers (less relevant as mostly supplied by other companies including Hitachi, Siemens Energy, GE Vernova).
- **Low Voltage Electrical:** Low voltage switchgear, LV transformers, protection equipment, busways.
- **Uninterruptible Power Supply (UPS):** Medium and low voltage UPS systems including batteries.
- **Rack PDUs:** Power strip/distribution device taking power from the upstream electrical system and distributes it to the IT equipment in the rack.
- **Cooling:** Solutions including traditional in-row or in-rack cooling, chillers but also liquid cooling solutions including CDUs. Vertiv has a large traditional (and liquid) cooling franchise while for the rest of the coverage, it's targeted on specific applications.
- **DC Infrastructure:** Racks, cable management, installation products.
- **Building Control:** Building management systems, HVAC, fire and security.
- **Software & Services:** Maintenance, datacenter infrastructure management software.
- **Distributed IT:** Single phase UPS, secure power not related to datacenters.

Figure 8: Datacenter revenues 2025 in US\$ bn



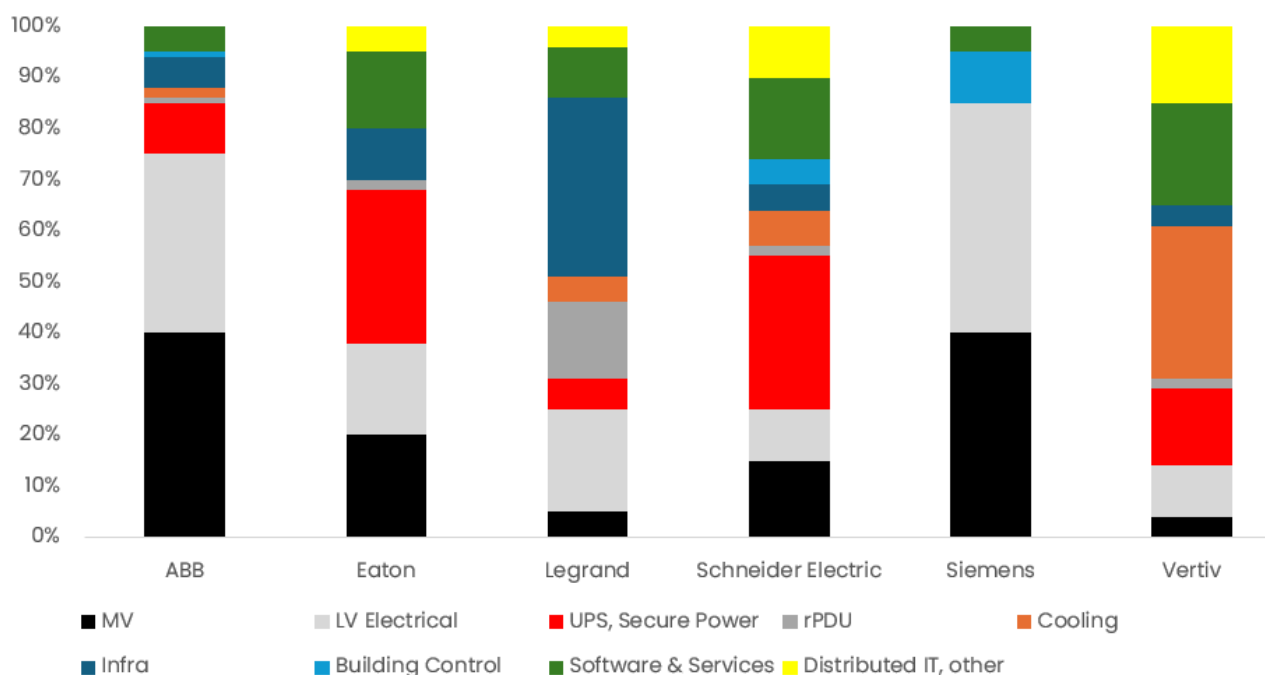
Source: Company data, Oxcap Analytics estimates. ABB excludes DC exposure in Motion (only includes Electrification). Siemens excludes DC exposure in Digital Industries (Control, Digital Twin etc). Includes distributed IT exposure. Eaton excludes Boyd Acquisition (2026 closure).

Figure 9: Share of Datacenter of 2025 group revenues



Source: Company data, Oxcap Analytics estimates. ABB excludes DC exposure in Motion (only includes Electrification). Siemens excludes DC exposure in Digital Industries (Control, Digital Twin etc). Includes distributed IT exposure.

Figure 10: Indicative estimate of DC exposure to product category (2025 revenues)



Source: Company data, Oxcap Analytics estimates based on management comments, industry reports and market share statistics.

Note: Distributed IT and other includes small UPS business. LV Electrical includes power electronics, LV transformers and busways. Schneider and Vertiv and to some extent Eaton have sizable distributed IT businesses selling secure power solutions to non-DC customers. The **Schneider** breakdown is adapted from the information provided at the Dec 2025 CMD, which is best-in-class. **Vertiv** disclosed a breakdown at its May 2026 CMD. **Siemens** is estimated based on disclosure at its Dec 2024 SI CMD which shows that DC at that time represented about 15% of each Electrification and Electrical Products revenues which we weigh by the size of the business with Electrification mostly related to MV but also some LV solutions while Electrical Products is mostly LV Solutions. **Legrand's** breakdown is estimated based on the group's communication at the Q4 2025 and Q1 2026 results calls and clarification provided by IR but as categories are bundled, our confidence in the numbers is lower. **ABB** does not provide a break-down of its datacenter exposure (despite its otherwise best-in-class disclosure approach) but commented that the biggest contributor to its data center revenues is the Smart Power division, followed by Distribution Solutions and (by some distance) Smart Buildings which allows for some approximation. **Eaton** does not provide a break-down of its datacenter business and hence represents our estimate of triangulating market positions from industry research reports and company commentary. It excludes the Boyd acquisition that closed in March 2026. The Boyd acquisition adds \$1.7 bn of annualized 2026E sales, according to Eaton, with 80% of it reflecting liquid cooling. On a pro-forma basis, Cooling would then represent ~15% of 2026 datacenter sales.

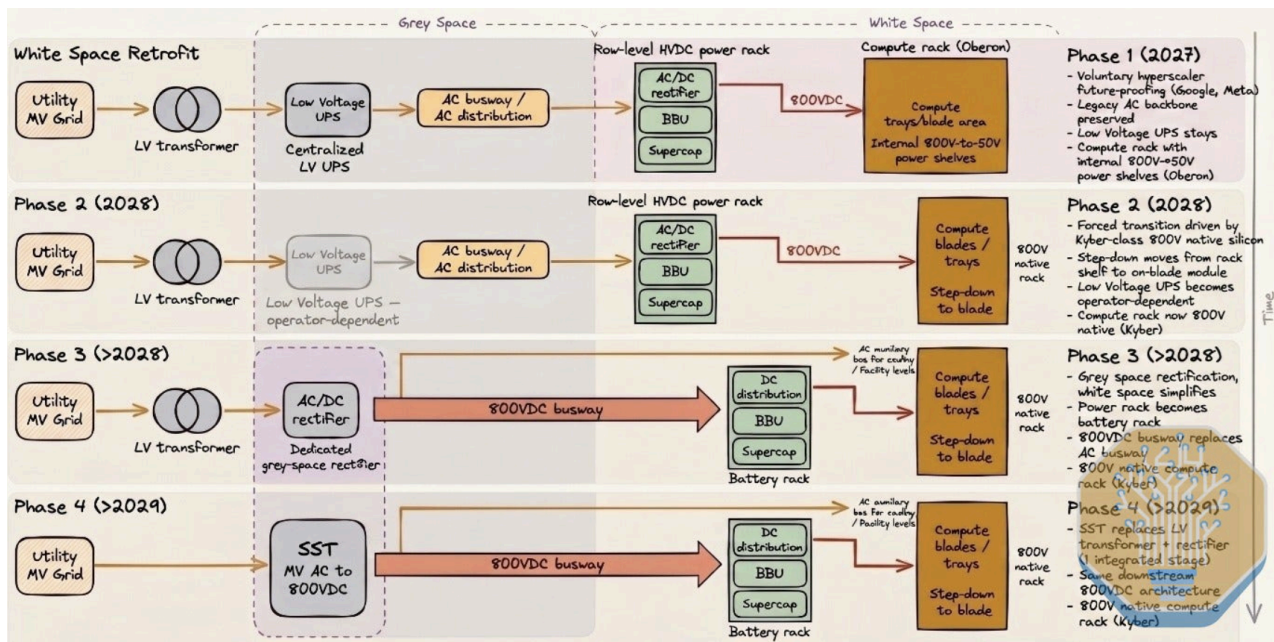
Primer on the 800VDC Transition and impact on suppliers and market opportunity

Figure 11: Transition Risk & Opportunities – a simple summary table

Transition	Timeline	What happens?	Where is the opportunity?	Where is the risk?
Phase 1	2027+	Traditional rack PDU's phased out	HVDC sidecar ABB, Eaton, Schneider, Vertiv, Delta	Traditional Rack PDU Legrand < 20% of portfolio
Phase 2	2028+	Low voltage UPS starting to be phased out	HVDC sidecar ABB, Eaton, Schneider, Vertiv, Delta	Low Voltage UPS Eaton, Schneider, Eaton Vertiv, lesser degree ABB and Legrand
Phase 3	2028-29	HVDC sidecar replaced by battery rack Addition of AC/DC rectifier in grey space	Battery pack, AC/DC rectifiers ABB, Eaton, Schneider, Vertiv, Delta incl others	HVDC sidecar, centralized UPS ABB, Eaton, Schneider, Vertiv, Delta
Phase 4	2029+	Solid state transition, AC architecture disappears	Solid State Transformers, Power Electronics ABB, Eaton, China XD, GEV, Hitachi, Delta among others DC busways and protection ABB, Eaton, Schneider, Legrand, Vertiv and others	Traditional Low Voltage AC architecture ABB, Eaton, Schneider, Legrand, and others

Source: Oxcap Analytics, based on SemiAnalysis data

Figure 12: Four Phases of technology transition



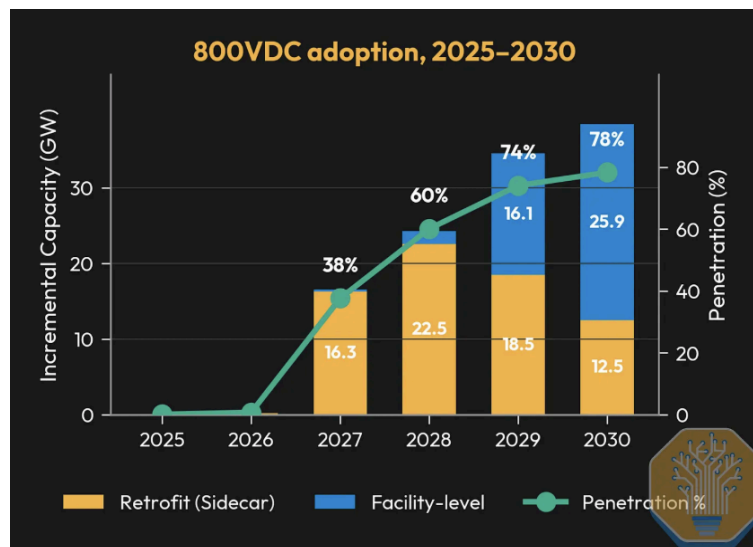
Source: SemiAnalysis, May 2026.

Figure 13: Change in market opportunity in \$ million per MW throughout the 800V DC transition

	Introduction	MV Transformers	MV Switchgear	Low voltage electrical equipment	UPS	AC-DC Rectifier	Solid State Transformer	Busway, Busbar	Rack PDU	HVDC Power Pack Sidecar	Battery Pack	Power Shelf/PSU	Total \$mn/MW	Change
Equipment spending per \$mn/MW														
Baseline \$ mn/MW	NM	0.35	0.70	0.55	1.20	0	0	0.30	0.05	0	0	0.20	3.35	
Phase 1 (The Sidecar)	2027	0.35	0.70	0.55	1.20	0	0	0.35	0	0.40	0	0.20	3.75	12%
Phase 2 (Goodbye UPS)	2028	0.35	0.70	0.55	0	0	0	0.35	0	0.40	0	0.30	2.65	-29%
Phase 3 (Centralisation)	2028/2029	0.35	0.70	0.55	0	0.20	0	0.35	0	0	0.20	0.30	2.65	0%
Phase 4 (The Destination)	>2029	0.35	0.70	0	0	0	1.25	0.25	0	0	0.20	0.30	3.05	15%

Source: SemiAnalysis, Oxcap Analytics. Excludes generator market which is included in SemiAnalysis market total as none of our companies supply them and their use is not impacted by the 800V DC transition. Generators faces challenges from trend to behind-the-meter power solutions, gas engines and battery systems which have not been reflected in the SemiAnalysis forecast of a steady market opportunity per MW for generators.

Figure 14: The real design changes start in 2029 onwards



Source: SemiAnalysis.

What Changes?

Phase 1 of the 800V DC transition: additive for electrical suppliers, with Rack PDUs the exception

Timing and drivers. Phase 1 (2027 onwards) is driven more by customer preference and future-proofing than by technical necessity. First-generation NVIDIA Vera Rubin systems can still be supported by traditional three-phase AC, so adoption will vary by customer, density target and deployment philosophy.

What changes and what does not. The upstream architecture remains familiar: medium-voltage power is stepped down through transformers, passes through double-conversion UPS systems and is distributed as 415V AC via busbar to the IT rows. The change occurs at the rack interface. Instead of feeding 415V AC into each IT rack, the feed is routed into a dedicated HVDC power rack ("sidecar") next to the compute racks. The sidecar has three functions:

- AC-DC conversion from 415V AC to 800V DC
- short-duration backup / ride-through via a battery backup unit
- transient-load management, using capacitors or other buffering to absorb rapid GPU load swings

Inside the IT rack, a DC-DC power shelf steps 800V DC down to around 50V DC for the compute trays, and on-board voltage regulator modules perform the final conversion to the sub-1V levels required by CPUs, GPUs and accelerators.

Economics: additive, not substitutive. The sidecar adds conversion, protection, control, energy-buffering and mechanical integration content, which more than offsets the removal of the rack PDU. SemiAnalysis estimates that electrical spend per MW rises by around 12% (Figure 13). The sidecar represents a \$0.4mn per MW opportunity, peaking at around \$11bn in 2028 before declining as later phases move the DC architecture upstream. It is therefore probably a temporary market.

Who benefits. We expect Schneider, Vertiv, Eaton, Delta Electronics and ABB, among others, to target the sidecar with products currently under development. The product combines rectifier modules, power semiconductors, a battery backup unit and potentially supercapacitors, using a mix of in-house and sourced components.

Who is exposed: Legrand. Traditional Rack PDUs are phased out. This is a small market (~2% of electrical spend ex generators, according to SemiAnalysis) but relevant for the market leader Legrand, where it may account for around 15% of revenues (Oxcap estimate, not confirmed by the company). Schneider, Vertiv, Eaton and ABB also sell Rack PDUs, but their exposure is materially smaller and likely more than offset by the sidecar opportunity.

Legrand addressed the topic in the Q&A of its Q4 2025 and Q1 2026 results:

- Around 20% of revenues, including Rack PDUs and UPS, could be negatively affected by 800V DC, while the impact on the remaining ~80% is neutral to positive (we believe in absolute terms, not relative to competitors).
- Management expects this to be offset by growth from rising rack power draw and size, the sidecar cooling opportunity, and new products such as DC busways.
- The company points to its record of adaptation: a datacenter business built from scratch since 2017 through more than 30 acquisitions and organic expansion.

Phase 2 (2028 onwards): 800V DC goes rack-native and the central UPS starts to lose its role

Driver: rack density, not preference. Phase 1 is a transitional architecture around NVIDIA's Oberon rack, inserting a sidecar while leaving the upstream AC infrastructure intact. Phase 2 is driven by the power density requirements of the Kyber rack design, which makes the move to 800V DC less discretionary.

What changes. 800V DC is distributed deeper into the compute rack. Rather than stepping down to around 50V DC in a centralised rack-level power shelf, conversion moves to on-blade or near-blade power modules close to the load. The rest of the architecture initially remains in place.

Consequence: the low-voltage UPS becomes dispensable. A traditional UPS converts AC to DC to charge a battery and back to AC to maintain power quality during outages. Neither step is needed in a DC architecture, where battery backup units integrated at the rack or sidecar can take over ride-through and transient-load management. SemiAnalysis expects the centralised low-voltage UPS to lose relevance progressively and eventually become obsolete. The transition will not be uniform:

- Some operators will remove the UPS altogether. Google and Meta already run designs without one, using distributed backup to improve efficiency and cut conversion losses.
- Some will retain centralised low-voltage UPS for redundancy, operational familiarity or load-fluctuation management, at least in the medium term.
- Others will move to medium-voltage UPS at the grid or campus connection point, such as ABB's HiPerGuard.

Economics: a (relative) shrinking addressable market for traditional suppliers. UPS systems represent about one third of datacenter electrical content in current designs. They are the traditional core of the datacenter business at Schneider and Eaton and are also important to Vertiv and others. Where the central UPS is removed, electrical cost per MW falls materially and efficiency improves, although part of the saving is offset by higher spending on rack- and server-level power electronics, including batteries.

Where the value goes. Value shifts away from UPS equipment towards high-voltage DC distribution, rack-level energy storage, protection, controls and power supply units. The PSU and power-module opportunity is likely to accrue primarily to electronics and power-conversion specialists such as Delta, Lite-On and Flex Power, rather than to traditional electrical equipment vendors.

Phase 3 (2028/2029): Centralisation. 800V DC becomes the backbone, with value redistributed rather than lost

What changes. In Phases 1 and 2, AC-DC conversion sits close to the IT load in the HVDC sidecar. In Phase 3, rectification moves upstream into the grey space or outdoors, creating an 800V DC backbone that feeds rows, racks and energy-buffering systems directly. The sidecar is replaced by a separate AC-DC rectifier higher up in the stack and a battery pack.

What stays. Medium-voltage transformers and switchgear remain essential because the utility feed and most on-site generation are still AC (fuel cells, which generate DC natively, are a partial exception).

What goes and what replaces it. The low-voltage AC layer between the rectifier and the IT load is materially simplified:

- Removed or reduced: low-voltage switchgear, centralised UPS systems, floor PDUs and parts of the AC busway.
- Added: DC busway, battery or energy racks, DC/DC distribution units, BBU shelves and potentially supercapacitors to manage fast GPU load transients.
- New protection layer: DC breakers, DC switchgear and solid-state circuit breakers.

Economics: similar TAM, different mix. The total addressable market per MW is broadly similar to the prior phase, and the vendors are likely to be largely the same. Lost low-voltage AC content is partly replaced by new DC distribution, protection and control content. Value moves away from the integrated sidecar and the conventional low-voltage AC chain towards separate rectifiers, battery racks, DC busway, DC protection and rack-level power electronics.

Vendor positioning. Suppliers are already moving in this direction. Delta is positioning a full-stack 800V DC ecosystem, including solid-state transformers, DC busways, DC/DC power shelves and liquid cooling. ABB is emphasising medium-voltage protection and UPS architectures such as HiPerGuard. SemiAnalysis expects other electrical suppliers, including Legrand, to follow.

Phase 4 (after 2029): Solid-state transformers. The clean-sheet design, with value shifting to power electronics

What changes. A solid-state transformer (SST) converts medium-voltage AC from the grid or behind-the-meter generation directly into 800V DC, which feeds a battery pack near the rack and then straight through to the compute trays. The existing low-voltage AC infrastructure disappears. (Fuel cells, which generate DC natively, do not need the conversion step.)

Why SemiAnalysis calls it the "holy grail". A conventional design steps medium voltage down through an iron-and-copper transformer and then rectifies low-voltage AC into DC. An SST does both in one stage using high-frequency semiconductor switching, typically based on SiC MOSFETs. The benefits are:

- a footprint reduction of potentially up to 90%
- better efficiency, controllability and dynamic response
- programmability: SSTs can adjust under load, support bidirectional power flow and potentially allow datacenters to use their batteries as grid-support assets during system stress

The drawback is maturity. SSTs are new to mission-critical datacenter applications, with limited operating history, open questions on useful life and reliability, and additional cooling and protection requirements.

Economics: early, but TAM-accretive. SemiAnalysis estimates that SSTs remove roughly \$0.75mn per MW of low-voltage equipment and rectifier cost, against an initial SST cost of \$1.0–1.5mn per MW, which should decline as the technology scales. It does not expect broad adoption before 2029, but believes major hyperscalers are already running pilots and sees the SST market reaching around \$13bn by 2030.

Who captures the value. An SST is built from electronic components rather than copper and steel, so it is a fundamentally different product from a traditional transformer. The incremental opportunity may therefore not accrue to the low-voltage incumbents but to a broader field. We discuss the SST opportunity later on page 24.

Facility level load still requires AC

Even in an 800V DC architecture, datacenters do not become fully DC facilities. A significant portion of the non-IT load — including chillers, pumps, fans, lighting, general HVAC and building controls — will continue to rely on AC power. CDUs are generally AC-powered today, although Delta has disclosed a DC-powered CDU and suppliers are also working on DC-based compressor technologies for chillers. The likely outcome is therefore a hybrid architecture: DC increasingly dominates the IT power train, while AC remains relevant for much of the facility infrastructure.

Backup power is also unlikely to disappear. The move to 800V DC may accelerate the use of supercapacitors, battery back-up units and battery energy storage for transient-load management, ride-through and short-duration backup. However, longer-duration outages remain difficult to cover economically with batteries alone, given the size and cost of the required battery overbuild relative to generator or engine based solutions. In our view, generators, fuel cells or other long-duration backup technologies will therefore remain relevant, even as short-duration resilience migrates closer to the rack.

The key investment point is that DC architectures do not eliminate the electrical equipment opportunity; they broaden it. As datacenters become a larger share of the overall electrical market, especially in the US, the incentive to develop DC-specific components increases. This should support new addressable markets in DC busway, DC protection, DC-powered cooling, rectification, BBUs, supercapacitors and rack-level power electronics.

Challenges of the transition

The NEC is the core US framework for electrical installation, but approval is ultimately down to local authorities. The 2029 NEC cycle is expected to provide a more complete treatment of 800V DC data center architectures; before then, deployments are likely to require site-specific approval and equipment-level certification. That favours hyperscalers with internal engineering resources and repeatable campus designs, while raising the adoption hurdle for smaller operators and colocation providers.

The safety challenge is that DC fault interruption is fundamentally different from AC. AC voltage crosses zero 100 or 120 times per second in 50Hz and 60Hz systems, creating natural arc-extinction points. DC has no such zero-crossing, so arcs can be more persistent and harder to interrupt. Traditional AC protection cannot simply be reused; 800V DC architectures require DC-rated protection, isolation, fault detection and potentially solid-state breakers. This also raises the labour requirement: in a market already short of qualified electrical technicians, high-voltage DC adds another layer of training, certification and operational complexity.

Medium voltage UPS opportunity

An MV UPS is one of three ways to solve the 800V HD design problem. Hence it is not a mandatory component and straight replacement of the low voltage UPS revenue. The design choices of the hyperscalers will define the outcome.

The 800V DC design converts utility/BTM medium voltage once, at the perimeter of the hall, to 800V DC, distributes that on a DC busway and steps it down once more at the rack level to the voltage required by the chips. The classic chain of MV/LV transformer, LV UPS, AC busway and PDU and in-rack rectification to DC disappears.

What does not disappear is the need for ride-through energy storage and control (to provide immediate back up power until the generator / alternatives kick in) and fault protection.

In the 2027-2028 hybrid architecture discussed earlier which uses a sidecar/power rack, keeps the LV UPS in place. Hence the transition to MV UPS (and SST) likely accelerates in 2028 onwards.

The UPS function can be put in three different places:

- **Medium Voltage UPS:** A static UPS at the medium voltage level secures the plant AC power and feeds it to large rectifiers that create the 800V DC bus. The advantage is that the product is available today and tested in commercial applications. It is also technology agnostic to what happens after this step as it can either feed a traditional AC set-up or a AC/DC rectifier. This is where the ABB HiPerGuard plays.
- **Medium Voltage DC UPS:** This is a solution that combines a MV/LV transformer, a AC/DC rectifier and a battery on the DC side. This merges the UPS and the rectifier into one objects with the battery on the 800V DC bus. This approach is popular in China / Huawei approach.
- **Solid state transformer:** This solution is often described as the end-game and the theoretically cleanest solution but with implementation challenges given the technology is not commercially available and the supply chain not ready yet. Costs are likely higher, initially. A medium voltage solid state transformer subsumes the transformer, rectifier and UPS into one box. The MV UPS or MV DC UPS would be a bridge product for 2027-2029 until SSTs are ready.

Revenue impact: Can sales of MV UPS replace the lost revenues for the current LV UPS suppliers (mainly Schneider, Eaton and Vertiv)? Only partially, and likely not in the long term. Why only partially:

1. Consolidation: One 25 megawatt medium voltage UPS (10x 2.5MW units stacked) replaces 1-2MW LV UPSs, resulting in a lower \$/MW despite the increase in the cost per unit.
2. Migration: Much of the value moves into power semiconductors, DC protection and batteries.
3. Longer term SST replacement: The degree of adoption of SST is highly uncertain at this point as it depends on the cost curve, performance and durability of a new product.
4. Competition: The UPS market is relatively consolidated with Schneider, Eaton and Vertiv the global big 3, with ABB, Delta and Legrand smaller. We estimate UPS to be a profitable business for the market leaders, given their dominant position and scale. A new product allows for new market entrants and likely results in ramp-up costs and higher R&D.

Vendor overview:

- **ABB HiPerGuard:** ABB is the early market leader with commercially deployed MV UPS in the field ([LINK](#)). ABB has a relatively small position in traditional UPS and hence has likely more to gain than lose, at least prior to a potential full move to SST.
- **Vertiv:** the company is a first mover with its MV DC UPS offering available from H2 2026.
- **Eaton:** The company is not offering a static MV UPS and instead bets directly on the end point of a medium voltage solid state transformer, based on its acquisition of SST start up Resilient Power.
- **Schneider:** No MV UPS or SST announced yet
- **Siemens:** No MV UPS, focus on SST development with partner Reinhausen.
- A number of transformer manufacturers have announced SST development plans (see below).
- **GE Vernova:** Not a major player in MV/LV datacenter electrical solutions but have introduced a MV UPS solution.
- Start-ups: A number of start ups offer MV UPS, including Ominverter and ON.Energy.

Solid State Transformers – a new product category

SSTs represent one technology architecture to address the need for higher powered compute in the medium term. Solid state transformers are also of interest as a technology solution for other direct current applications, including electrical vehicle charging, battery storage and solar applications.

Solid State Transformers are not necessary to adopt a 800V DC datacenter architecture but provide potentially a superior solution to the power supply architecture. A SST replaces the traditional passive MV to LV stepdown iron transformer with a power-electronics device that can take medium voltage AC and convert it straight to a regulated DC bus, feeding the batter pack und ultimately the servers.

The advantages are:

- **Fewer conversion stages** resulting in higher end-to-end efficiency, resulting in lower power losses and hence higher PUE for the datacenter.
- Higher DC voltage means lower current for the same power, which results in **material savings** on copper and other materials used for busbars.
- Power density steps up because the space and size of the power magnetics run at kilohertz, rather than 50 or 60 Hertz run in traditional applications collapses. This allows for **more space for compute**, particularly relevant when retrofitting old datacenter shells for modern AI compute.
- **Active controls and grid services** – a game changer longer term. As a power electronics product, rather than a passive iron transformer, a SST can do voltage transition, power-factor correction, reactive power support, harmonic filtering and fault-current limiting and it's inherently bidirectional. This matters because grid interconnection has become the binding constraint on datacenter buildout. A facility that can actively support the grid, ride through disturbances, and provide flexibility is easier to interconnect and can participate in demand-response and grid-stability markets — turning the power system from pure cost center into a source of optionality.
- **Handling volatile AI loads:** Large training clusters create sharp, synchronized power swings as tens of thousands of GPUs step load together. SSTs' fast control loops can buffer and smooth these, protecting both the utility connection and internal equipment in ways a passive transformer cannot. This can reduce the need for battery storage or other buffers to handle the load fluctuations.
- **Energy-router role for on-site generation and storage.** As datacenters pair with batteries, fuel cells, on-site power, a multi-port SST can act as the microgrid hub — marrying MV AC, DC sources, and storage in one controllable device. It can also absorb functions that today sit in a separate UPS layer, potentially eliminating redundant conversion and equipment.
- **Standard designs:** Traditional transformers can't be customized once purchased, hence each product needs to be engineered to specification. This is somewhat different in China, which has a modern grid based on standard products, and to some extent for distribution transformers for large datacenters where a large number of units with equal design can be purchased. SST can be build on standard components and then programmed for a specific application. This should allow to over time reduce the high cost of an SST when compared to traditional solutions.

What are the challenges of the transition to SST

Given the list of advantages of SST discussed above, why are they not used today and what are the hurdles for the adoption? SST are a new product category that is currently not commercially available. Traditional transformers are very good at their job - relatively cheap and durable and widely available. While lead-times for large power transformers can stretch to the end of the decade, the transformers that could be replaced by solid state alternatives (MV step-down / low voltage transformers) generally have shorter lead times with a larger set of suppliers and shorter lead-time for adding capacities.

- **Traditional transformers are very good at their one job:** A traditional transformer is extremely durable and reliable due to the lack of active components, often with a 30-40 year life. Efficiency is as high as 99% and maintenance costs are close to zero. "Boring and proven" is exactly what utilities, but also datacenter owners want. Challenging a product that until recently at least, was cheap and available and has a 99% energy efficiency and last 40 years is a tall order.
- **The biggest challenge is proven reliability:** A SST swaps a simple passive device (copper, steel and insulation material) for hundreds or thousands of active semiconductor switches, gate drivers, capacitors and control, with each potentially representing a critical failure point. Capacitors have a limited life-time, as have required cooling fans. Redundancy can address some of the issues but at an incremental costs and space requirement.
- **Upfront costs:** Upfront capex costs for SST are materially higher than iron transformers with the economics only justified by system level benefits, at times harder to underwrite than stick price comparisons. Silicon Carbide (SiC) and Gallium Nitride (GaN) semicon costs need to decline and production volumes have to ramp to improve the cost equation. According to Mordor Intelligence, will cost around 10x the cost of a conventional transformer.
- **Supply chain scaling:** SST need SiC or GaN power devices. Wafer yields are improving but higher voltage SiC modules are still relatively immature with unclear long term reliability under high loads and high temperatures.
- **Efficiency and cooling:** The reduction in conversion stages in a datacenter improved energy efficiency but the SST itself has a lower efficiency than an iron transformer as hot semiconductors need active cooling, adding a moving part and coolant dependency.
- **Lack of standards:** There is no mature UL (US) or IEEE standard for SST or trained field workforce qualified or experienced to work with SST. A full DC datacenters needs the broad ecosystems to move and adopt standards in order for the SST transition to happen. SST will likely lag the overall DC adoption in datacenters. The NEC 2029 code will likely include 800VDC code support but datacenters coming online prior to that will need facility level approval.

SST Market Segmentation

The SST market can be segmented by product type:

- Distribution SST: Used mainly for grid-edge distribution, the largest end-market
- Traction SST: rail traction transformers, a niche market
- Power SST: DC microgrid deployment

It is also important to note that SST are often classified into Low Voltage, Medium Voltage and High Voltage but that these voltages are not comparable to the scale used for iron transformer. A high voltage SST may have a >35kV rating, while a large power transformer used in a high voltage transmission setting may have a 765kV rating.

SST adoption in datacenters

Based on the SemiAnalysis 800V DC framework the adoption of solid state transformer technologies would only start towards the end of this decade, as the final step in their transition model to a full DC native architecture.

Solid State Transformers may not turn out to be the best technology to electrify datacenters of the future. The move to a DC being used in AI datacenter architecture is pretty certain, given the involvement of the chip systems by NVIDIA and others. There is no technical requirement to use SST for datacenter power architecture, hence they need to win on cost, performance and durability.

The alternative technology is the continued use of a centralized AC/DC rectifier solution, described earlier in the report as the Phase 3 of the higher voltage DC transition according to the SemiAnalysis framework. Here, Utility or behind-the-meter power is stepped down using a traditional low voltage transformer to a AC/DC rectifier sitting in the grey space of the datacenter from where a DC busway feeds a battery back close to the servers.

The chart below compares the two designs.

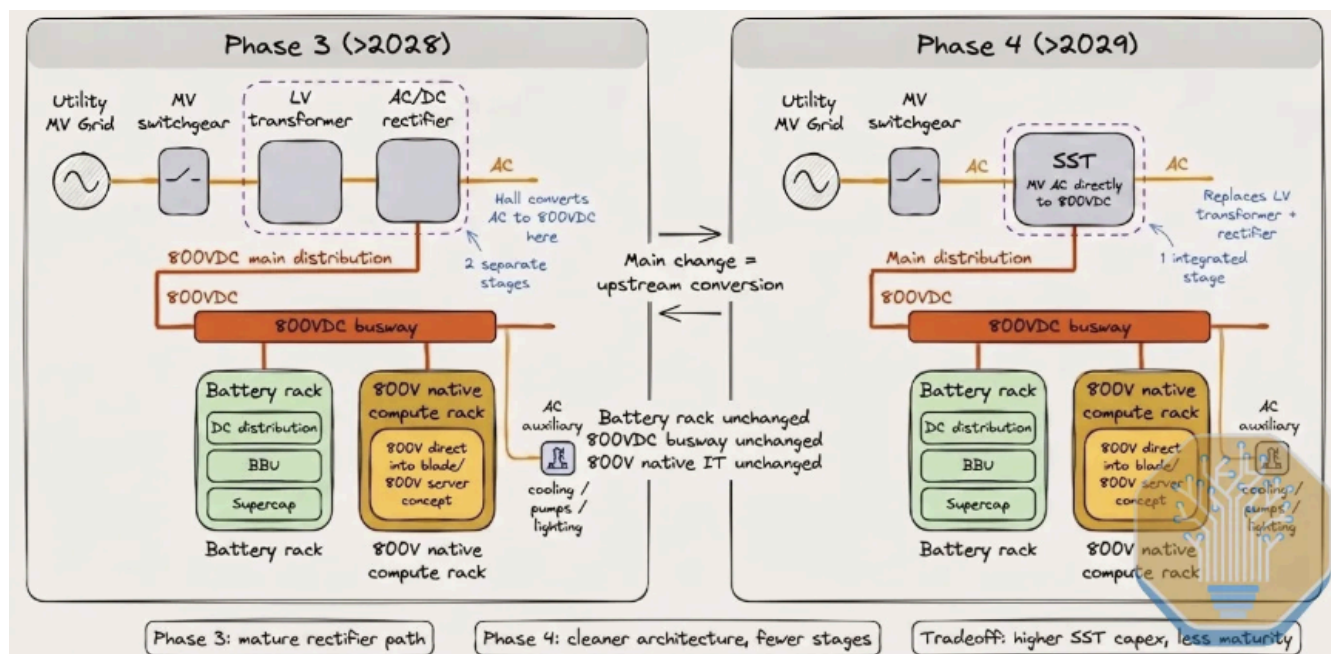
The main attraction of using SST instead of a rectifier/LV transformer design is higher energy efficiency, translating into opex savings for the operator and unlocking higher compute capacity. SemiAnalysis mentions an overall system efficiency improvement from around 82-85% to 97%, driven by eliminating two conversion steps. State of the art SSTs currently can achieve a 98% efficiency with an engineering target to get this to 99%. Different vendors now state >98% efficiency levels but have generally yet to prove this at larger sizes and under continuous load.

Overall, datacenter efficiency should improve for a full DC power design with SSTs when compared to today's architecture. This is mainly because the elimination of the conversion in UPS (~3% loss) and in the Power Supply Unit (PSU) that sits close to the chips and converts incoming AC current into low voltage DC (~6% loss). In a full DC architecture, datacenter power efficiency could increase from 82% to 87%-88%. This would result in incremental 70MW of compute for a 1GW datacenter.

SSTs are also dramatically smaller than the conventional transformers used. A conventional transformer operates at 50 or 60 Hz and needs a massive iron core. An SST switches at 20,000 Hz or higher, shrinking the core by roughly 90%. That is where Infineon's claimed 40x weight reduction and 14x size reduction to come from.

Relevant to datacenters, SST are also programmable and bidirectional. This allows datacenters to play a more active role on the grid. SST designs may also be able to accommodate multiple power inputs (grid, behind-the-meter AC power, DC power from fuel cells or solar and battery storage systems).

Figure 15: Comparing the LV transformer and rectifier design versus SST for datacenters



Source: SemiAnalysis.

We expect first SST deployments to start in 2027 as trials and adaption to scale towards the end of the decade. The SST solution likely comes at a higher cost but is more compact and programmable.

Theoretically, there would also be the option to use SST at the higher voltage level, but the supply chain for the required SiC power electronics is unlikely economically viable in the medium term, with cost prohibitive, making low voltage SST an earlier-to-market technology. We don't expect SSTs to play a meaningful role in power grids overall in the next 5-10 years.

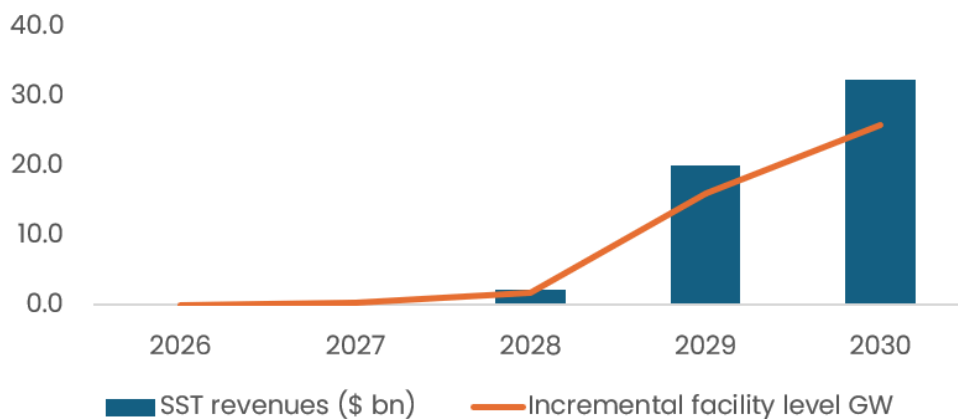
TAM for SST and what changes in datacenters

The SST replaces Low Voltage equipment of around \$0.55 million per MW and a rectifier of around \$0.2 million per MW. This translates into a revenue headwind for low voltage electrical suppliers of \$750 million per 1 GW of datacenter build out.

This compares to SemiAnalysis' estimate of initial SST costs of \$1 to 1.5mn per MW or a market opportunity of \$1 to 1.5bn of revenues per 1 GW of datacenter build out. Hence, the transition represent in a cost increase of ~\$0.5mn per MW, or corresponding increase in addressable market for suppliers. Given the early stage of the SST development, the cost per MW range is large, and speed of adoption will depend heavily on how fast the price could be reduced.

Semi-Analysis expects a datacenter SST market opportunity of \$2.2bn by 2028 and \$32.4bn by 2030, based on a market of 2GW in 2028 and 23GW in 2030.

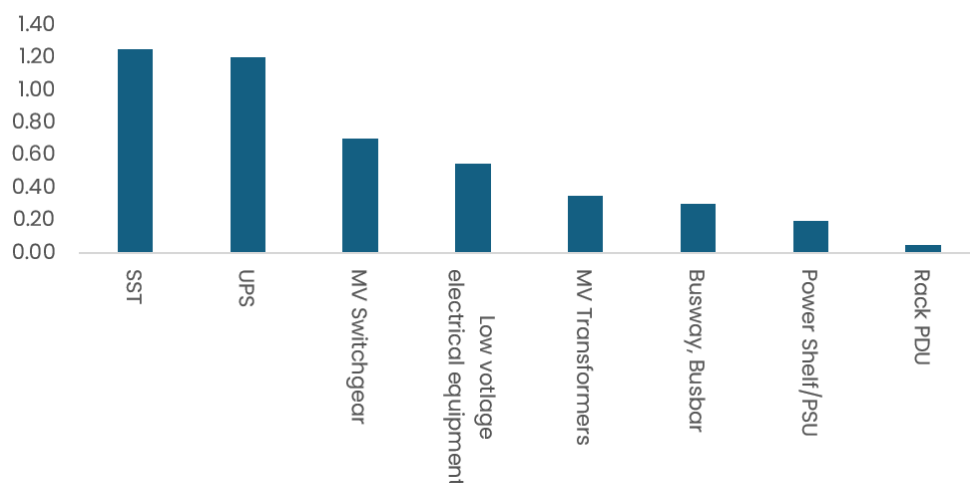
Figure 16: Datacenter market opportunity for SST



Source: SemiAnalysis.

In terms of value per MW for an AI datacenter, SST would be the largest category when compared to today's \$/MW electrical product categories. However, SST will likely only be used in a subset of datacenters, resulting in an overall smaller total market size when compared to UPS.

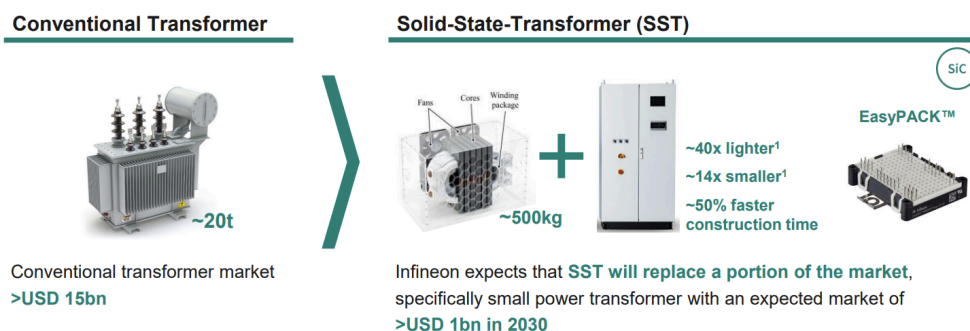
Figure 17: \$ / MW equipment datacenter equipment capex



Source: Analysis.

Forecasts for the size of the datacenter related SST market vary widely. For example, Infineon expects a SST market of >\$1bn by 2030, compared to SemiAnalysis' \$32bn. The uncertainty results form a number of variables. First, the total market for datacenters and share of AI datacenters that you use high performance GPUs that require a higher voltage DC architecture. Second, the cost of the SST per MW which should get cheaper over time as the technology matures and production values increase.

Figure 18: Infineon expects a SST market of >\$1bn by 2030



Source: Infineon.

Data on SST pricing per MW are scarce. No SST vendor publishes list prices yet, and most units shipping in 2026 are pilots priced case-by-case. Thunder Said Energy estimated a cost of 5-10x a traditional distribution transformer, which cost around 25,000-45,000 per MW. This would imply even at the high end of the range a price of only \$450,000 per MW, one third of the SemiAnalysis assumption. However, we are not sure the two estimates are comparable in terms of scope what is priced.

If a SST would only cost \$450,000 per MW than the cost advantage versus a traditional architecture would be material, given that the SST design would also get right of the need for LV switch gear, UPS conversion, PDUs and rectifier which cost ~\$1 mn per MW or more.

What drives the cost of a SST

The bill of materials is dominated by power electronics, not iron and copper. Roughly: power semiconductors (SiC MOSFETs) and their gate drivers ~30–35% — the single largest line item; the high-frequency transformer ~15–17%; DC bus capacitors ~15–17%; thermal management ~8–10%; filter inductors and sensing ~8–10%; control electronics ~7–9%; and auxiliary parts the remainder.

Unlike a conventional transformer whose cost is mostly commodity materials (copper, steel) and high and complex labor assembly costs with little room to fall, an SST's cost is mostly manufactured semiconductor content, which is the kind of cost that follows a learning curve and natural electronics deflation:

- SiC device prices have been falling ~10–15% a year as the industry moves to 200 mm wafers and EV-driven volume scales the supply chain.
- Higher-voltage devices (6.5 kV and 10 kV SiC MOSFETs, which Wolfspeed and others are commercializing) collapse the cell count — a 10 kV device does the work of four or five 2.3 kV cells, cutting drivers, capacitors, and assembly along with the silicon.
- Volume manufacturing replaces today's hand-built pilot units with automated production of standardized modular blocks (the Heron/Amperesand/SolarEdge designs are all explicitly modular 1–5 MW building blocks for this reason).
- Power-electronics products with similar structure (solar inverters, EV traction inverters) showed 15–20% learning rates.

Vendor analysis

We profile the existing SST vendors on page 36. Discussing market shares is not meaningful given most are at the prototype stage. Just because one datacenter supplier has already acquired an SST start up does not necessarily mean they will capture a disproportionate share of the SST opportunity.

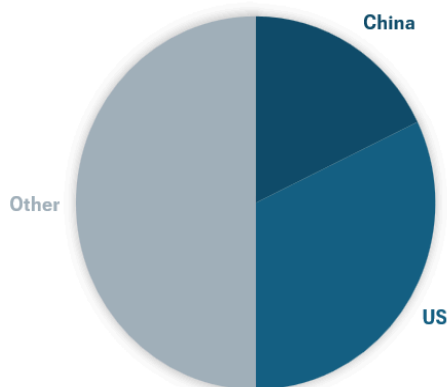
We categorise the impacted companies into five different groups:

- Vendors of medium voltage transformers and low voltage AC electrical equipment which may get replaced by SST / 800V DC technology in parts of the datacenter market.
- Suppliers of traditional transformers across broader end-markets.
- Start-ups developing SSTs
- Electronics suppliers entering the market
- EV-charging or solar inverter companies given shared technology bricks with SST

We count around 30 companies that have announced or discussed solid-state transformer developments. There are likely many more that we did not find information on, either because the development is in an early stage, or there is limited information on those companies, particularly from China and other Asia countries.

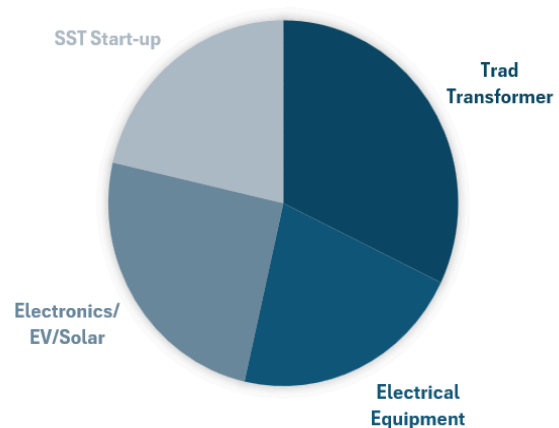
We count 9 traditional manufacturers of power or distribution transformers that have announced or talked about solid state transformer developments.

Figure 19: HQ location of companies developing SST



Source: Oxcap Analytics.

Figure 20: Type of company developing SST



Source: Oxcap Analytics.

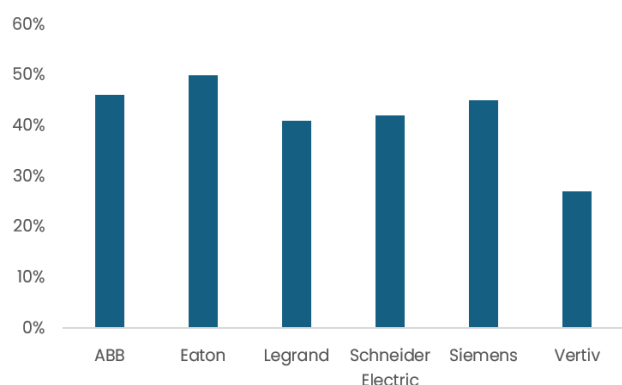
Datacenter electrical vendors

The potential shift to SST impacts the current electrical equipment vendors in two ways.

First, as suppliers of traditional low voltage transformers that get replaced. The value per MW for the MV to LV step-down transformer is relatively small compared to the overall typical electrical content per MW \$3-3.5mn (excluding generator, on-site power, range is large and depending on redundancy requirements and configuration). They are used to step down the load feeding the servers but also cooling systems and other auxiliary equipment. Hence traditional transformers may still be needed even in a full 800 DC architecture with SSTs, unless other equipment also transitions to DC supply.

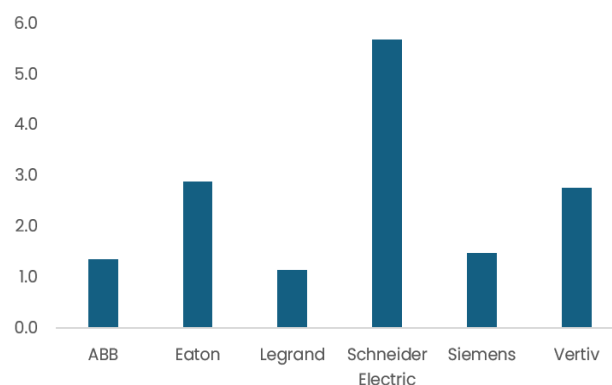
Second, they are also exposed to the wider architecture change impacting their low voltage products business, including UPS, more broadly, which accounts for the large share of their exposure to a change in the architecture as shown in the charts below.

Figure 21: Share of 2025 datacenter revenues impacted by DC architecture transition



Source: Company data, Oxcap Analytics. Note: only includes electrical revenues. Cooling industry is also going through changes. Percentage does not imply revenues at risk as only part of datacenter industry moves to a new direct current architecture.

Figure 22: \$ bn revenues exposed to datacenter architecture transition



Source: Company data, Oxcap Analytics. Note: only includes electrical revenues. Cooling industry is also going through changes. Percentage does not imply revenues at risk as only part of datacenter industry moves to a new direct current architecture.

Traditional transformer suppliers

Historically, the larger transformer businesses were owned by the diversified electrical engineering companies. ABB sold its business to Hitachi. Siemens spun off Siemens Energy and Alstom sold its Power Grid business to GE, which later spun off GE Vernova. In general, while Siemens and ABB kept their medium voltage and in the case of Siemens Grid Control/Smart Grid businesses, the the medium voltage transformer businesses went with the transmission to the new company. The MV/LV transformer business is hence part of Siemens Energy and Hitachi today, not their former parent company. Schneider and Eaton offer some transformers for low voltage applications but don't have a broad MV/LV transformer business.

Transformer revenues are not disclosed within the Grid businesses of the major suppliers but we estimate that Hitachi is the clear market leader, followed by Siemens Energy and then GE (including Prolec) in value terms while TBEA in China is the larger player in volume. Other major suppliers include SGB-SMIT, Mitsubishi Electric, Hyundai Electric, Hyosung Heavy, Toshiba and CG Power.

Entering the SST market mostly represents an incremental revenue opportunity for the transformer suppliers. While they may lose some demand for traditional distribution transformers, the increase in the value per MW represents a meaningful opportunity. The very large part of the global transformer market is unlikely replaced by SST given the higher costs and much shorter durability. We don't expect Power Transformers to see a meaningful impact, if any at all.

Among the major transformer companies, Hitachi, GE Vernova, Mitsubishi Electric and Hyosung have announced SST developments as have a number of Chinese suppliers. Interestingly, Siemens Energy has yet to announce a plan.

The traditional distribution transformer market is fragmented with many specialists and niche suppliers. In addition to the major Transformer companies discussed above, the main suppliers of MV/LV transformers for datacenters are Hammond Power Solutions, Virginia Transformers, ERMCO, Howard Industries, Forgent Power, Central Moloney, Maddox, WEG, Rex Power Magnetics and Olsun. In Europe R&S, Koncar, Best, Astor, TMC Transformers, Ormazabla, Wilson Power Solutions, Trafo Elettro and Efacec are supplying the market.

Entrants from the electronics industry

A SST consists mostly of electronic components, rather than copper, oil and steel and hence companies with strong electronic competences, supplier network and electronic assembly capabilities are likely well positioned to success in the SST industry. Delta Electronics (listed in Taiwan) is an early market leader. Enphase Energy, a leading US listed supplier of solar inverters, home battery storage systems and EV-chargers is targeting a full system demonstration by Q4 2026.

Start ups

A number of start-up firms have introduced prototypes or plans for SSTs. The list includes DG Matrix, Amperesand, Heron Power and Novos Power. Eaton acquired Resilient Power Systems in July 2026. Resilient Power Systems is expanding from EV charging solutions into SSTs which is now offered under the Eaton brand. ABB has a stake (size undisclosed) and partnership with DG Matrix while Siemens works with Reinhausen.

We see the potential for start-ups to be acquired by traditional transformer companies or electrical vendors to accelerate their time to market. While start-ups can offer fast technology development and differentiated solutions, we believe that most large buyers, particularly in the datacenter space, tend to buy solutions from established vendors with strong sourcing and production capabilities and the capital to scale production and support warranties. Customers tend to prefer vendors with broad based capabilities across the electrical products used in a datacenter.

Figure 23: Overview of companies offering or developing SST technology

Company	Country	Ownership	State of development	Size	Efficiency	Comment
ABB	Switzerland	Public	Potential development given patent filing, unconfirmed	NA	NA	Owens minority stake in DG Matrix (March 2025)
Maschinenfabrik Reinhausen	Germany/UK	Private	Under development, targeting DC market	Up to 36kV	NA	Partnership with Siemens for SST development
Ampersand	Singapore	VC	Pilot deployments 2022-2025, now also in DC plus first commercial deployment	10-35kV	>98.5%	30MW deployment plan for 2026, Raise ~\$100mn. CEO ex Tesla
China XD	China	Public/Gov controlled	Limited public information, likely leading Chinese player with commercial product	10kV / 2-3MVA	>97.5%	Speculated to work with Vertiv, deployed in Chinese grid
Delta Electronics	Taiwan	Public	Strong background in EV charging applications, some SST shipments to Chinese DCs	10-33kV (staked)	up to 98.5%	First trial shipments with US Datacenters
DG Matrix	US	VC backed, including ABB, MHI	First commercial shipment 2026 for solar/battery/grid. Offer grid, DC and charging solution	10-34.5kV	98.50%	Raised ~\$100mn in funds.
Eaton	US	Public	Product brochure of MVSSST available	15kV (35kV 2027)	>97%	Eaton acquired Resilient Power Systems in July 2025 for \$86mn
Enphase	US	Public	GaN-based, demos 2026, customer pilots 2027, volume shipments 2028.	15kV and 35kV, 2.5M	98.500%	Company focus is microinverters & batteries
Flex	US	Listed	SST Roadmap defined	NA	NA	Acquired EPC Power
GE Vernova	US	Public	Hyperscaler Pilot H2 2026, commercial sales 2027. 5MW prototype installed.	NA	NA	Based on results call comments GEV
Heron Power	US	VC backed	2026 pilot, 2027 deployment, Interact Power and Crusoe as customers	34.5kV, 4.2MW	98.50%	~\$0.2bn raised, CEO ex Tesla
Hitachi	Japan	Public	R&D focus for DC SST, no commercially, catalogued SST	NA	NA	ABB legacy of PETT SST for rail
Huawei	China	Public	Strategy and roadmap but no product details	NA	NA	Strong power electronics and integration capabilities
Hyosung Heavy	South Korea	Public	SST module demonstrated May 2026, earlier SSTs for grid distribution	23kV	NA	NA
HD Hyundai Electric	South Korea	Public	Ongoing development, limited detail	NA	NA	NA
Hyperscale Power	Switzerland	VC backed	No commercial product, focus on footprint, application target DC, Solar and EV charging	13.8kV / 3.3MW	98.50%	Differentiated architecture with high frequency and small size
Jinpan Technology	China/US	Public	Testing of product with customers	10kV/2.4MW	NA	US based business active in distribution transformers for DC
LS Electric	South Korea	Public	Focus on DC ecosystem for datacenter with SST and dedicated plant, Hyperscaler order	NA	NA	Strong DC technology focus
Mitsubishi Electric	Japan	Public	No product announcement or prototype but mentioned in market research reports	NA	NA	Company has in-house SiC capabilities and could be supplier
Schneider Electric	France	Public	No clear SST strategy or development announced.	NA	NA	NA
Siemens	Germany	Public	August 2026 partnership to work with Maschinenfabrik Reinhausen/Amatys, Product TBC	up to 36 kV	NA	Amatys Power (see above)
Sifang Shuzhi / Digital Intelligence	China	Public/ Gov controlled	Commercial SST offering	10-20kV	>98%	35kV offering to be launched later in 2026
SolarEdge	US	Public	2026 lab demos, 2027 launch, 2028 deployment ramp	13.8-34.5kV	>99%	Historic focus on solar inverters
Sungrow	China	Public	EnerNeo SST, Chinese leader with domestic datacenter sales	3MW, 10-14 kV	98.50%	Background in inverters/electronics
TBEA	China	Public	Still in R&D phase, according to analyst comments in China	NA	NA	NA
Toshiba	Japan	Public	No product announcement yet but company has many of the technologies in-house	NA	NA	NA
Transforma Energy	US	Private	R&D phase, focus on utility distribution grid. Claim to have demonstration product	NA	NA	Very limited details available
Vertiv	US	Public	In development	35kV at 3-5MW	NA	Rating is targeted industry typical solution
WatEV	US	VC backed	No prototype announced. Focus on fast charging but also AI datacenter	NA	NA	Core focus is commercial vehicle charging

Source: Company websites, trade information, Claude AI.

SST are a national priority for China

In March 2026, the Chinese MIIT, NDRC, SASAC and the National Energy Administration jointly issues the *Implementation Plan for High-Quality Development of Energy-Savings Equipment (2026-2028)*. The policy calls for the development of large-capacity solid state transformers as part of its overall policy effort to increase the share of highly-efficient transmission and distribution systems. The support is mostly driven by soft factors, not through purchase mandates or an SST specific quota or target. We could not find evidence of a SST specific subsidy. China XD is leading the standard setting with the industry. There is a scheme that supports buyers of new technology in this space with insurance to reduce the risk for the customers. To date, the deployment mostly relates to pilot projects for demonstration of the technology.

Technology and product overview

A solid-state transformer (SST) — also called a power electronic transformer or smart transformer — is best understood as a transformer that has been reinvented as an active power-electronics system rather than a passive electrical product of copper and iron. It does the same headline job as a conventional transformer (change voltage levels, provide isolation) but replaces the big 50/60 Hz magnetic core with high-frequency switching and a much smaller high-frequency transformer, and in doing so gains a lot of intelligence and controllability that an ordinary transformer simply cannot have.

On the technology, the core trick is frequency. A conventional transformer is large and heavy because it operates at grid frequency (50/60 Hz), and the required amount of iron and copper scales inversely with frequency. An SST first uses power semiconductors to convert the incoming AC into a high-frequency waveform — typically tens of kilohertz — passes it through a compact high-frequency magnetic core for voltage change and galvanic isolation, then converts it back to whatever the load needs.

A typical architecture has three stages: an AC-to-DC rectifier on the medium-voltage input side, an isolated DC-to-DC stage built around that high-frequency transformer, and a final DC-to-AC (or DC-to-DC) stage feeding the load. Because the medium-voltage side has to withstand thousands of volts, the input stage is usually built from many switching cells stacked in series. The whole thing is enabled by wide-bandgap semiconductors — principally silicon carbide (SiC), and increasingly gallium nitride at lower voltages — which switch fast at high voltage and temperature with low losses; SiC is really the ingredient that made SSTs practical rather than just a lab curiosity.

What you get for that added complexity is a device that is far more than a voltage changer. Because power flows through controllable electronics, an SST can regulate voltage precisely, correct power factor, filter harmonics, and support the grid dynamically — functions that normally require separate equipment. It exposes one or more DC ports natively, which matters enormously as the world adds solar, batteries, EV chargers and DC-hungry data-center loads, since it avoids redundant AC-DC-AC conversions. It provides fault isolation and can limit fault currents. And it is dramatically smaller and lighter for a given rating — often cited as a fraction of the size and weight of an equivalent conventional unit — which is valuable where space is expensive.

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